

A Appendix

A.1 The Determination of the Space Factor ζ

When the leaf nodes of LIFT trigger a SMO, the expanded space $n.size_{new}$ follows Equation 1.

$$n.size_{new} = \zeta \cdot \min\left(1, \frac{n.insert_rate_t}{n.insert_rate_{t-1}}\right) \cdot n.size_{prev} \quad (1)$$

For the determination of the space factor ζ , see Equation 2.

$$\zeta = \begin{cases} \eta, & n.size_{prev} \geq 1M \\ 3\eta, & n.size_{prev} \geq 500K \\ 6\eta, & n.size_{prev} < 100K \end{cases} \quad (2)$$

As indicated by Equation 2, nodes of varying sizes should be equipped with corresponding space factors. Equation 1 demonstrates that smaller nodes require larger space factors to ensure adequate space reservation. For example, suppose two hot nodes have slot capacities of 8 and 16, respectively; they would require expansion factors of 6 and 3 to achieve a total expansion of 48 units. The space base factor η can be dynamically adjusted based on different workloads.

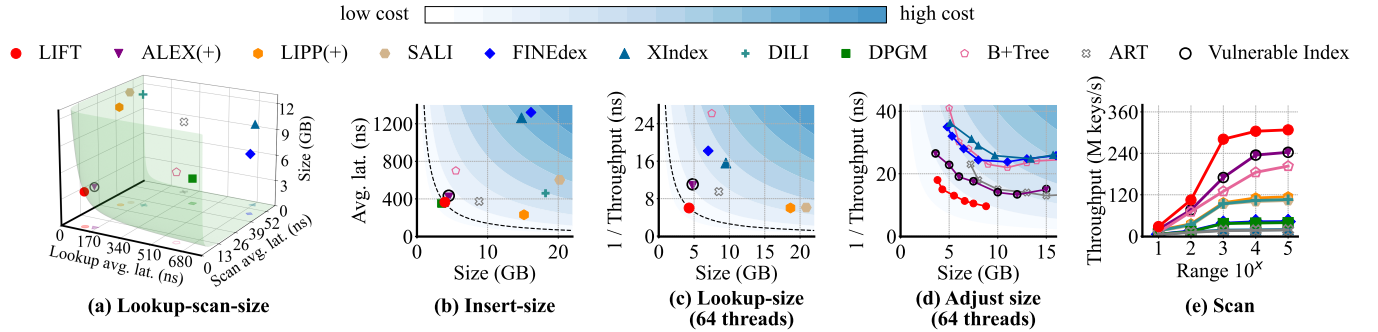


Figure 1: Comparison with ALEX and other indexes on the Libio dataset.

A.2 Comparative Evaluation with ALEX

Figure 1 presents a comparison of LIFT with ALEX and other indexes under different workloads. As shown in Figure 1, LIFT consistently outperforms ALEX across various workloads, achieving an optimal time-space tradeoff and surpassing ALEX in range query performance as well. Therefore, LIFT not only demonstrates strong robustness but also exceeds the previously best-

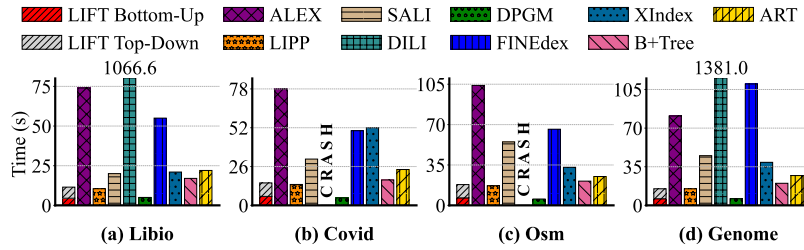


Figure 2: Bulk load time comparison with ALEX and others.

Figure 2 shows the loading times for ten indexes when performing bulk load of 200M keys across four datasets. Here, LIFT Bottom-Up denotes the construction time of Algorithm 1 (bottom-up), while LIFT Top-Down denotes the construction time of Algorithm 2 (top-down). Due to the need for cost model calculations, ALEX has significantly higher loading times compared to most other indexes. DPGM, which constructs the structure recursively from bottom to top using the PLA algorithm, has the shortest loading time. In contrast, LIFT requires interpolation to calculate the optimal error ϵ and density d , resulting in longer loading times than DPGM. LIPP has a loading time comparable to LIFT, and is notably faster than ALEX.

A.3 Additional Experimental Results

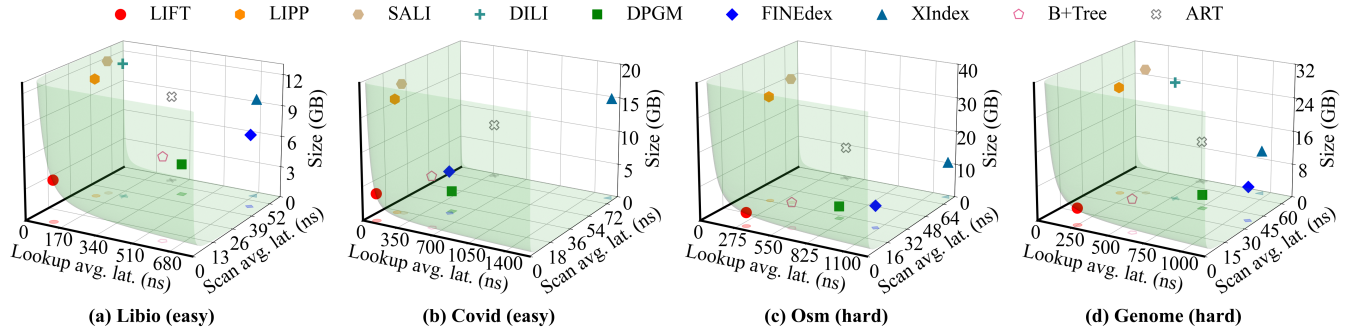


Figure 3: Lookup-scan-size tradeoff evaluation on the Libio, Covid, Osm, and Genome datasets.

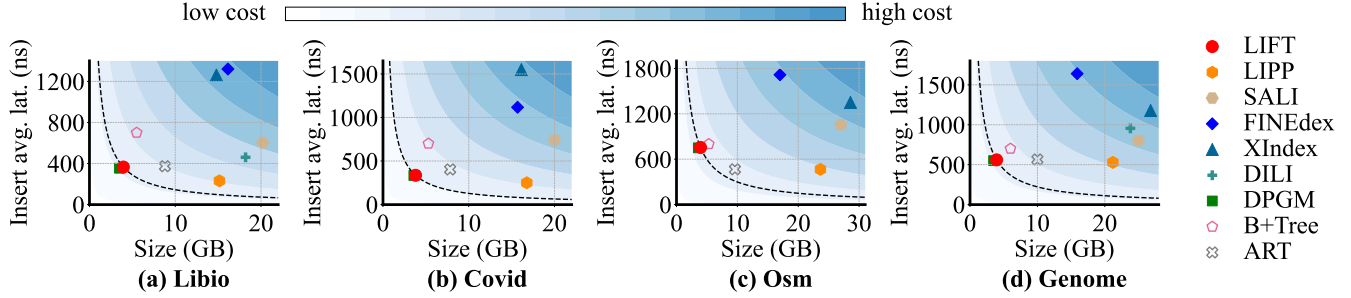


Figure 4: Insert-size tradeoff evaluation on the Libio, Covid, Osm, and Genome datasets.

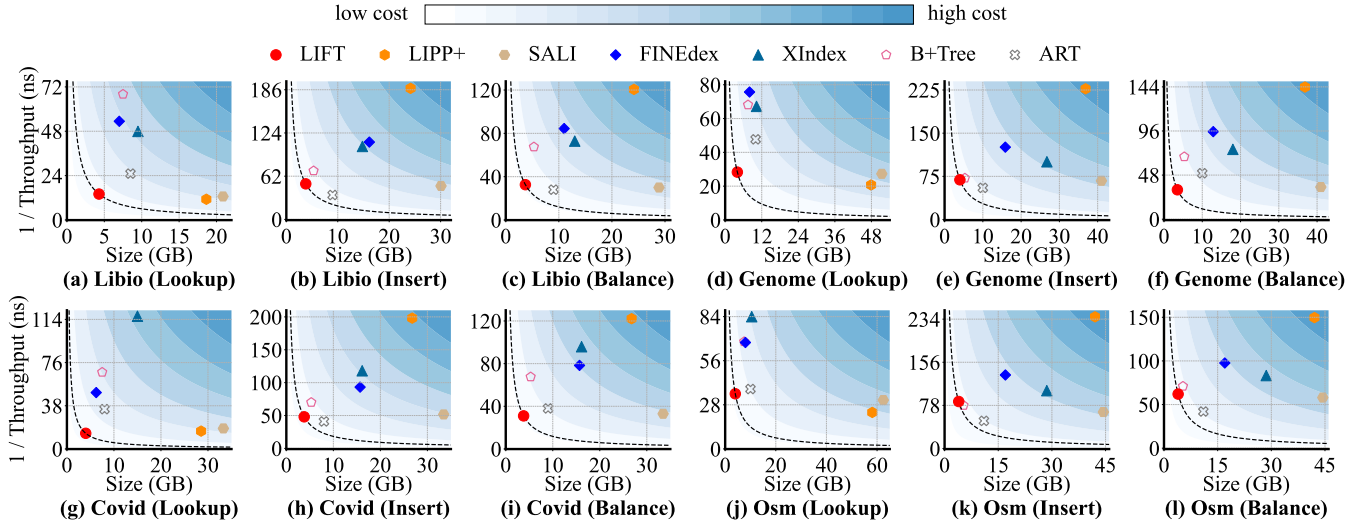


Figure 5: Time-space tradeoff evaluation with 16 threads on the Libio, Covid, Osm, and Genome datasets.

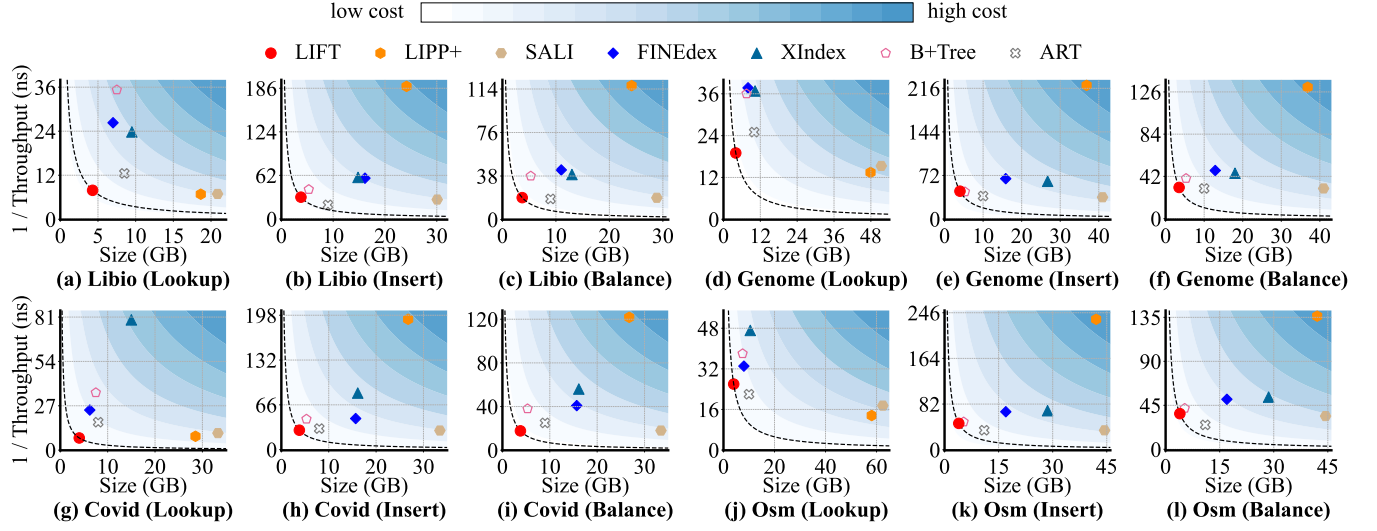


Figure 6: Time-space tradeoff evaluation with 32 threads on the Libio, Covid, Osm, and Genome datasets.

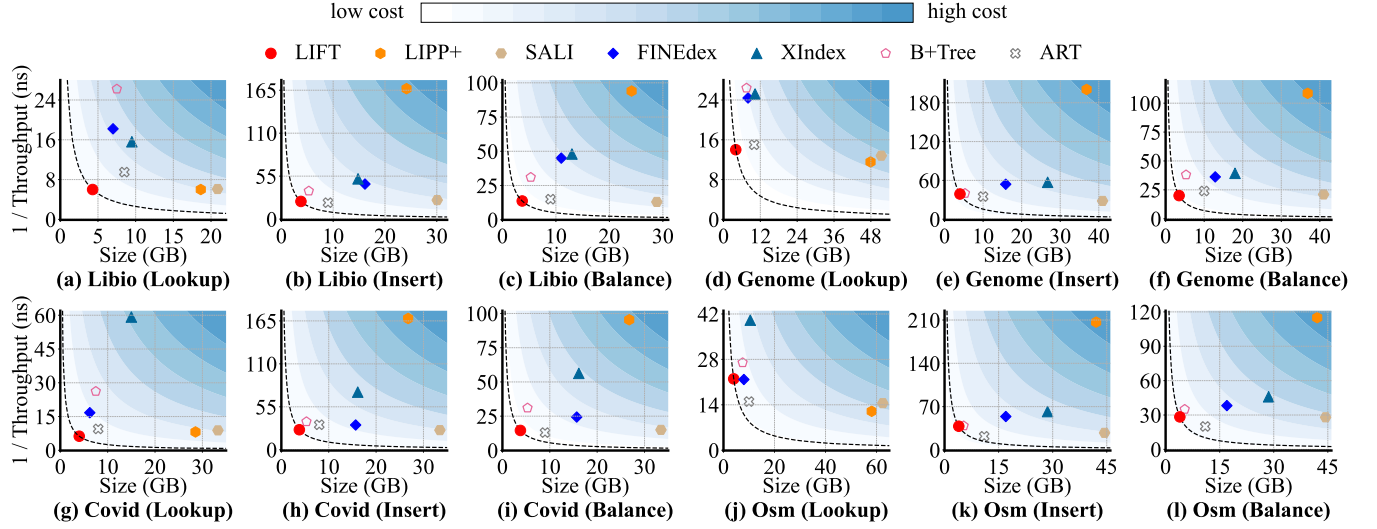


Figure 7: Time-space tradeoff evaluation with 64 threads on the Libio, Covid, Osm, and Genome datasets.